INES ALONSO

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Research interests: *stochastic dynamics of molecular and learning systems; Markov state models and kinetic analysis; verifiable machine learning for the physical sciences*

EDUCATION

Florida Polytechnic University — *Lakeland, FL, USA*
B.S. Computer Science, Minor in Applied Mathematics · GPA 3.95/4.0 — **Expected Dec 2026**

University of Barcelona — *Barcelona, Spain*
Undergraduate physics coursework (three semesters); transferred to Florida Polytechnic — **Transferred 2024**

Relevant coursework: Machine Learning, Natural Language Processing, Quantum Information & Computing, Numerical Methods, Computational Linear Algebra, Algorithm Design & Analysis, Data Structures, Computer Architecture, Multivariable Calculus.

RESEARCH EXPERIENCE

Institute of Photonic Sciences (ICFO) — *Barcelona, Spain*
Research Intern in Computational Biophysics, SPIE@ICFO María Yzuel Fellow — **May 2025 – Present**
Supervisors: Prof. Nicoletta Liguori (Photon Harvesting in Plants and Biomolecules) and Prof. Maciej Lewenstein (Quantum Optics Theory). Question: does pH, or the pH sensor PsbS, change how the light-harvesting complex LHCII moves during photoprotection?

- Designed and ran the full kinetic analysis of **175 coarse-grained (Martini 2.2) LHCII trajectories, 4,500 µs of sampling** across three pH/PsbS conditions (MDTraj, deeptime, PyEMMA).
- Built an **out-of-core pipeline** streaming TICA covariance estimates over all **26,796 inter-residue distances** (~750,000 frames, ~80 GB) with no downsampling or feature pruning; trained a five-layer **VAMPnet** (PyTorch) to learn nonlinear slow coordinates.
- Built and stress-tested **Markov state models two independent ways** (clustering-based and physics-defined states) with Chapman–Kolmogorov tests, spectral-gap analysis, and held-out cross-replica validation.
- Designed the **statistical controls**: an identical-copy known-zero test, false-positive calibration over ~200 randomized comparisons, and power analysis.
- Compared results against the experimental literature and wrote the final report with proposed next simulations; manuscript in preparation.

Florida Polytechnic University, Physics Department — *Lakeland, FL*
Quantum Software Research Assistant · Advisor: Dr. Bradford A. Barker — **Feb 2024 – Present**
Collaborator: Dr. David A. Strubbe (UC Merced). Noiseless statevector benchmarks against FCI and CASCI for H_2, H_4, LiH, H_2O, and O_2.

- First author and co-developer of **q-sc-SF-EOM**, a quantum self-consistent spin-flip equation-of-motion algorithm that obtains ground and excited states from one eigenvalue problem built on a high-spin reference (Qiskit, PySCF, ADAPT-VQE).
- Reduced circuit cost at equal accuracy: in (2e,2o) active spaces (H_2, LiH, H_2O, O_2) the costliest matrix element drops from **52 to 16 CNOTs** (mean 21.3 → 12.5); for LiH, a spin-completed pool keeps excitation energies **within 0.1 eV of FCI at 3.3× fewer CNOTs** than conventional q-sc-EOM.
- Developed a **completeness analysis** of how much of the $M_S = 0$ determinant space each operator pool reaches (100% in (2e,2o), 40% for LiH (2e,5o), 25% for H_4), explaining when and why single flips yield spin-contaminated states across five molecules.
- Developed the excitation-operator selection scheme, including removal of operators that leave the target spin sector before diagonalization.

INDEPENDENT RESEARCH

Agentic MSM Benchmark Engine · github.com/inesalonsoo/multiagent_for2Dphysics — **Jul 2026 – Present**

- Built a **three-agent LLM system** (Orchestrator / Optimizer / Validator) that proposes Markov state model analyses of stochastic simulations; results are accepted only by **deterministic checks against exact physics**, never by the LLM.
- Validated the MSM engine against the **exact relaxation rate** of a stochastic double well (computed from the discretized generator): two metastable states found in **42/42 replicas**; measured/exact rate **0.975–1.031 for β = 3–7**.
- **Audited my own pipeline:** the textbook Eyring–Kramers gate is **6–10% off** at these temperatures, so the loop had accepted biased analyses; replaced it with an exact oracle and reported overconfident Bayesian error bars (**2 of 42** contain the true rate).
- Formalized the physics ground truth in **Lean 4 (Mathlib)**: 7 statements about the double-well potential (symmetry, derivative, barrier height), 2 proved so far, one with Axiomatic AI's **ax-prover**; proofs axiom-checked, and a test keeps the Lean file consistent with the Python code.

PUBLICATIONS & PRESENTATIONS

- **I. Alonso**, W. J. Hall, S. Mandulapalli, D. A. Strubbe, B. A. Barker, "The q-sc-SF-EOM quantum algorithm for ground- and excited-states in molecules and solids," manuscript in preparation.
- Poster, q-sc-SF-EOM, IEEE Quantum Week (QCE26), Toronto, Canada, Sep 2026.
- Poster and Lightning Round podium presentation (selected, 20 of 80 posters), NanoFlorida 2026 International Conference, Lakeland, FL, Oct 2026.
- Selected poster, STEM Research Competition (Undergraduate), SHPE National Convention, Indianapolis, IN, Oct 2026.

FELLOWSHIPS & AWARDS

SPIE@ICFO María Yzuel Fellowship, ICFO: competitive 12-month research fellowship 2025 – 2026
NanoFlorida 2026 Poster Competition: selected for the Lightning Round podium presentations (20 of 80 posters) Oct 2026
SHPE STEM Research Competition, Undergraduate Poster: selected; attendance fully funded Oct 2026
Florida Polytechnic University Merit Scholarship: $6,150 per semester 2024 – 2026
Esperanza Inc. Scholarship: $500 per semester for two semesters, with leadership training
President's List, Florida Polytechnic University Spring 2024, Fall 2024, Spring 2025
Quantum Excellence Badge, IBM Qiskit Global Summer School 2025

ADDITIONAL PROJECTS

Hierarchical Reasoning Model with Attention Feedback **Nov 2025**

- Designed the token-level module of a 60.6M-parameter Hierarchical Reasoning Model variant for multi-hop QA on HotpotQA, in which sentence- and token-level modules exchange information through bidirectional cross-attention (4-person team, NLP course, PyTorch).

Research Lead, Improved Tools for Mobile-Based 3D Scene Capture (Threshold360) **Sep 2025 – May 2026**

- Led a distributed 3D reconstruction pipeline (FastAPI, CUDA GPUs, Gaussian Splatting, Nerfstudio) generating virtual tours from mobile captures in under 60 minutes; built a React Native client for batched image upload over asynchronous REST APIs.

Space Trek Program, NASA Florida Space Grant Consortium, Center for Space Education **Mar 2025**

- Designed an altitude-triggered parachute deployment for a hydrogen balloon payload (Arduino, BMP280, MOSFET) and real-time acquisition of magnetic-field, cosmic-ray, and solar-UV data at altitude (Geiger counter, UV sensor, IMU).

LEADERSHIP & SERVICE

Sustainability President, Student Government Association, Florida Polytechnic **May 2025 – Jul 2026**

- Founded the university's Sustainability Agency; led 8 ambassadors, built a 100 m² campus garden on a $2,500 budget, and organized events with 200+ attendees alongside Residence Life and sponsors.
- Transitioned the role to a successor at the end of the term.

Vice President of Internal Affairs, Society of Hispanic Professional Engineers (SHPE) **Jan 2025 – Jul 2026**

- Ran 10 professional-development workshops and 8 heritage events; liaison to the Provost's Office and Career Services. SHPE 2025 Certified Chapter Leader.

Resident Assistant, Florida Polytechnic University **Aug 2024 – May 2026**

- Planned and ran 32 community events for a residential community of 1,000+ students.

TRAINING

IBM Qiskit Global Summer School 2025 · Barcelona Supercomputing Center CUDA Programming Course · SHPE Certified Chapter Leader 2025

SKILLS

Programming: Python, C/C++, SQL, Lean 4, LaTeX, Git/GitHub, Linux, Docker, REST APIs, Pydantic
ML & scientific: PyTorch, scikit-learn, deeptime, PyEMMA, MDTraj, NumPy/SciPy, Qiskit, PySCF, CUDA, MATLAB, LLM APIs
Methods: Markov state models, TICA, VAMPnets, coarse-grained MD analysis, stochastic differential equations, variational quantum algorithms, theorem formalization (Lean 4)
Languages: Spanish (native), Catalan (native), English (fluent), French (intermediate)